SESHADRIPURAM COLLEGE, BENGALURU

(Affiliated to Dr. Manmohan Singh Bengaluru City University)

DEPARTMENT OF COMPUTER APPLICATIONS (BCA)

PROJECT SYNOPSIS

"AI SMART CAMPUS INTELLIGENCE SYSTEM"

Submitted in partial fulfilment of the requirements for the award of the degree of BACHELOR OF COMPUTER APPLICATIONS (BCA)

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1. INTRODUCTION

A Smart Campus applies digital technologies to modernise how an institution manages academic operations, replacing physical registers and scattered spreadsheets with a centralised, real-time platform. Educational institutions generate continuous data — attendance, marks, and feedback — that is rarely analysed in time to help students who are falling behind. The AI Smart Campus Intelligence System addresses this by consolidating student data into a single web application and applying automated, explainable logic to flag academic risk early.

Existing manual systems suffer from disconnected records, delayed identification of at-risk students, no systematic feedback analysis, and no visual reporting. The proposed system solves this through a Flask web application backed by MySQL, offering role-based dashboards for Students, Faculty, and Admins, automated risk/placement scoring, Chart.js visual analytics, and sentiment-tagged feedback.

2. PROBLEM STATEMENT

Institutions rely on manual or semi-digital methods to track attendance, marks, and feedback, making it difficult to identify at-risk students early or correlate attendance with performance. There is a need for a centralised system that consolidates this data and applies a transparent, explainable evaluation logic to flag risk and readiness for every student.

3. OBJECTIVES

- Build a centralised web platform to manage student attendance, marks, and academic performance.
- Implement a rule-based engine to classify failure risk and placement readiness.
- Provide secure, role-based dashboards for Student, Faculty, and Admin users.
- Visualise performance data using interactive charts for quick interpretation.
- Capture and automatically classify student feedback sentiment using NLP.

4. SCOPE

The system covers secure login and role-based dashboards; storage of attendance and subject marks (DBMS, Python, AI); automatic computation of risk classification, placement estimate, and study suggestions; feedback submission with sentiment analysis; and Chart.js-based visual analytics. It is scoped for single-institution deployment with manually entered data; integration with external ERP systems or biometric hardware is left for future work.

5. EXISTING vs. PROPOSED SYSTEM

Existing System	Proposed System
Manual registers/spreadsheets; fragmented records	Centralised MySQL database for all student data
At-risk students identified only after results	Automated High/Moderate/Low risk flagging in real time
No systematic feedback analysis	Feedback auto-classified as Positive/Neutral/Negative
No visual performance trends	Chart.js bar/line charts for marks and attendance trends

6. LITERATURE SURVEY

- Educational Data Mining: studies show attendance and internal marks are strong predictors of final performance, supporting the threshold-based risk logic used here (Romero & Ventura, 2010).
- Sentiment Analysis: lexicon-based polarity scoring (as used by TextBlob) is an established method for classifying open-text feedback at scale (Pang & Lee, 2008).
- Learning Analytics Dashboards: visual, chart-based reporting helps students and faculty interpret trends faster than raw tabular data.

7. SYSTEM ARCHITECTURE & MODULES

The system follows a three-tier architecture. The Client Layer (HTML, CSS, JavaScript, Chart.js) renders role-based dashboards. The Flask Backend handles routing, authentication, and business logic, and invokes a rule-based evaluation module that compares attendance and marks against fixed thresholds (e.g., attendance below 60% or marks below 40% triggers "High Risk"). The MySQL Database Layer stores users, students, faculty, and feedback records.

Key Modules

- Authentication Module – verifies credentials and routes users by role (student/faculty/admin).
- Student Module – shows attendance, marks, risk/placement status, charts, and a feedback form.
- Faculty Module – lists students and allows profile management.
- Admin Module – consolidated view of any student's marks, attendance, and feedback, with edit access.
- Prediction Module – rule-based risk, placement, and study-suggestion logic (model.py).
- Feedback & Sentiment Module – captures feedback and classifies sentiment using TextBlob.

8. TECHNOLOGY STACK

Layer	Technology
Backend	Python, Flask
Frontend	HTML, CSS, JavaScript, Chart.js
Database	MySQL (via mysql-connector-python)
NLP / Sentiment	TextBlob

9. FEASIBILITY STUDY

- Technical: built entirely on open-source, well-documented tools (Python, Flask, MySQL, TextBlob, Chart.js) that run on standard hardware.
- Economic: no licensing costs; deployable on existing college lab infrastructure or low-cost hosting.
- Operational: simple, role-based dashboards require minimal training; explainable logic builds staff trust in the output.

10. FUTURE ENHANCEMENTS

- Upgrade the rule-based logic to a trained ML model (e.g., Random Forest) once sufficient historical data is available.
- Add password hashing and stronger authentication.
- Automated email alerts for High Risk students; QR/biometric attendance capture.

11. CONCLUSION

The AI Smart Campus Intelligence System delivers a working web application that centralises student academic data, flags at-risk students early through transparent rule-based logic, visualises performance trends, and captures sentiment-tagged feedback — giving students, faculty, and administrators a single, reliable source of truth for academic monitoring within the institution.

12. REFERENCES

[1] M. Romero and S. Ventura, "Educational Data Mining: A Review of the State of the Art," IEEE Trans. Systems, Man, and Cybernetics, vol. 40, no. 6, pp. 601–618, 2010.

[2] B. Pang and L. Lee, "Opinion Mining and Sentiment Analysis," Foundations and Trends in Information Retrieval, vol. 2, no. 1–2, pp. 1–135, 2008.

[3] M. Grinberg, Flask Web Development: Developing Web Applications with Python, 2nd ed. Sebastopol, CA: O'Reilly Media, 2018.

[4] Oracle Corporation, "MySQL 8.0 Reference Manual," [Online]. Available: https://dev.mysql.com/doc/refman/8.0/en/.

[5] Chart.js Contributors, "Chart.js Documentation," [Online]. Available: https://www.chartjs.org/docs/latest/.